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NYASA AGRICULTURAL OPTIMIZATION SYSTEM

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Abstract

Agriculture forms the foundation of Malawi's economy, supporting over 61.7% of the population and providing livelihoods for rural communities [1]. However, smallholder farmers face significant challenges including unreliable weather information, limited market access, poor record-keeping practices, and lack of intelligent decision-support tools [2]. These challenges result in low productivity, inefficient resource use, and continued food insecurity.

The Nyasa Agricultural Optimization System (NAOS) will be a web-based and mobile application platform designed to address these challenges. The system will provide farmers with real-time weather forecasts, market intelligence, digital farm record-keeping, and AI-powered crop recommendations [4]. By integrating modern technology into traditional farming practices, NAOS will enable farmers to make informed decisions, optimize resource usage, and adopt sustainable farming methods [10].

This Software Requirements Specification document outlines the functional and non-functional requirements for NAOS. It defines the system architecture, user interfaces, data flow, and technical specifications needed to develop a solution that will enhance agricultural productivity, improve farmer incomes, and contribute to food security in Malawi. The document serves as a blueprint for system development and establishes the foundation for subsequent design and implementation phases[9].

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1.0 Introduction

The Nyasa Agricultural Optimization System (NAOS) is a digital platform that will transform how smallholder farmers in Malawi access information and manage their farming operations. The system will integrate weather forecasting, market intelligence, digital record-keeping, and crop advisory services into a single, user-friendly platform accessible through both web and mobile interfaces.

NAOS operates in an environment where most farmers rely on traditional farming methods with limited access to modern technology. The system will bridge this gap by providing timely, localized, and actionable information that farmers can use to improve their decision-making processes [6]. The platform will support multiple user types including farmers, agricultural extension officers, buyers, and system administrators, each with role-specific functionalities.

The system will employ a modular software architecture, enabling independent development and testing of its core components. Each module, which are weather advisory, market intelligence and digital farm diary, will be designed as a distinct unit with well defined interfaces [4]. This architectural approach facilitates maintainability, scalability, and future enhancements. Additionally, the platform will be developed with a strong emphasis on user experience, ensuring simplicity and accessibility to accommodate users with varying levels of digital literacy [2].

NAOS will be developed following Agile methodology, enabling iterative progress and continuous stakeholder feedback [9]. This approach ensures that the system remains responsive to user needs and can adapt to changing agricultural conditions. The platform aims to shift smallholder farming from reactive to proactive, empowering farmers not just to survive but to thrive in a changing agricultural landscape [5].

2.0 Problem Statement

Smallholder farmers in Malawi face persistent challenges due to the fragmented nature of available agricultural information. Farmers rely on generalized radio weather forecasts and word-of-mouth for market prices, leading to unreliable planning and poor market timing. The lack of digital record-keeping tools results in lost or inconsistent farm data, making it difficult to analyze past performance. Without intelligent decision-support systems, farmers depend on traditional knowledge and memory, which often results in inefficient resource use and reduced crop yields. This cycle of inadequate information and outdated practices promotes low agricultural productivity, poor incomes, and continued food insecurity [1], [2], [7].

3.0 Scope of the Development Project

3.1 Features Included

The Nyasa Agricultural Optimization System will include the following features:

1. Weather Advisory:

- Real-time weather forecasts for specific locations
- Rainfall predictions and seasonal weather patterns
- Weather-based planting recommendations and optimal planting schedules
- Personalized crop suggestions based on location and weather patterns [4]
- Alert notifications for extreme weather conditions
- Historical weather data access
- Input optimization advice based on weather and crop type
- Adaptive learning from user feedback and farming outcomes

2. Market Intelligence:

- Daily crop price updates from local and regional markets [7]
- Price trend analysis and prediction (short term)
- Buyer directory with contact information
- Market demand information

3. Digital Farm Diary:

- Crop activity logging (planting)
- Harvest recording and yield documentation
- Income and expense tracking
- Historical performance analysis

4. User Management:

- Secure user registration and authentication
- Role-based access control
- User profile management

3.2 Features Excluded

The following features will not be included in the initial system release:

- Direct financial transactions or mobile money integration

- Loan application and credit assessment features
- Equipment rental or sharing services
- Satellite imagery analysis and IoT sensor integration [3]

3.3 Operational Environment Before System Installation

Before NAOS is installed, farmers rely on fragmented information sources. They access general weather forecasts from radio broadcasts and market prices through word-of-mouth or physical market visits, which is inefficient and leads to poor decision-making [2], [7]. Record-keeping is primarily manual or non-existent, making it difficult to analyze past performance or plan effectively [5]. There is no integrated digital platform providing personalized, data-driven advisory services to smallholder farmers in Malawi [9].

3.4 Operational Environment After System Installation

After NAOS is deployed, farmers will have centralized access to real-time, location-specific weather data and market intelligence, enabling proactive planning and better market timing [6]. The digital farm diary will facilitate evidence-based farming by allowing farmers to track inputs, monitor yields, and evaluate outcomes [5]. The AI-powered recommendation engine will provide personalized advice, helping to optimize resource use and improve productivity [4]. The system will interface with existing weather APIs and market data feeds without replacing them, creating a cohesive digital ecosystem that enhances, rather than disrupts, existing agricultural support structures [10].

4.0 System Personnel

4.1 System End Users

Farmers: The primary users who will consume weather, market, and advisory information and log data in the farm diary. They typically have small land holdings and varying levels of education and digital literacy [1].

Extension Officers: Secondary users who will monitor farmer activities within their zone, access aggregated data for trends, and use the platform to provide targeted advice and support.

Buyers: External users who access the platform to view produce listings and connect with farmers, improving market linkages [7].

System Administrators: Users responsible for managing user accounts, updating buyer directories, monitoring system health, and curating content.

4.2 Project Team

Student (Albert Msiska): Responsible for the overall project execution, including requirement analysis, system design, front-end and back-end development, database design and administration, API integration, quality assurance, and implementation of the AI recommendation logic. Manages project timelines and deliverables in consultation with the supervisor.

Supervisor (Tawonga Mkandawire): Provides overall project guidance, reviews progress, ensures academic rigor, facilitates access to resources, and validates the system against the project's objectives and scope.

4.3 Owner

The project owner is the student developer, Albert Msiska, under the academic supervision of Tawonga Mkandawire, representing the Department of Computer Engineering at the University of Livingstonia.

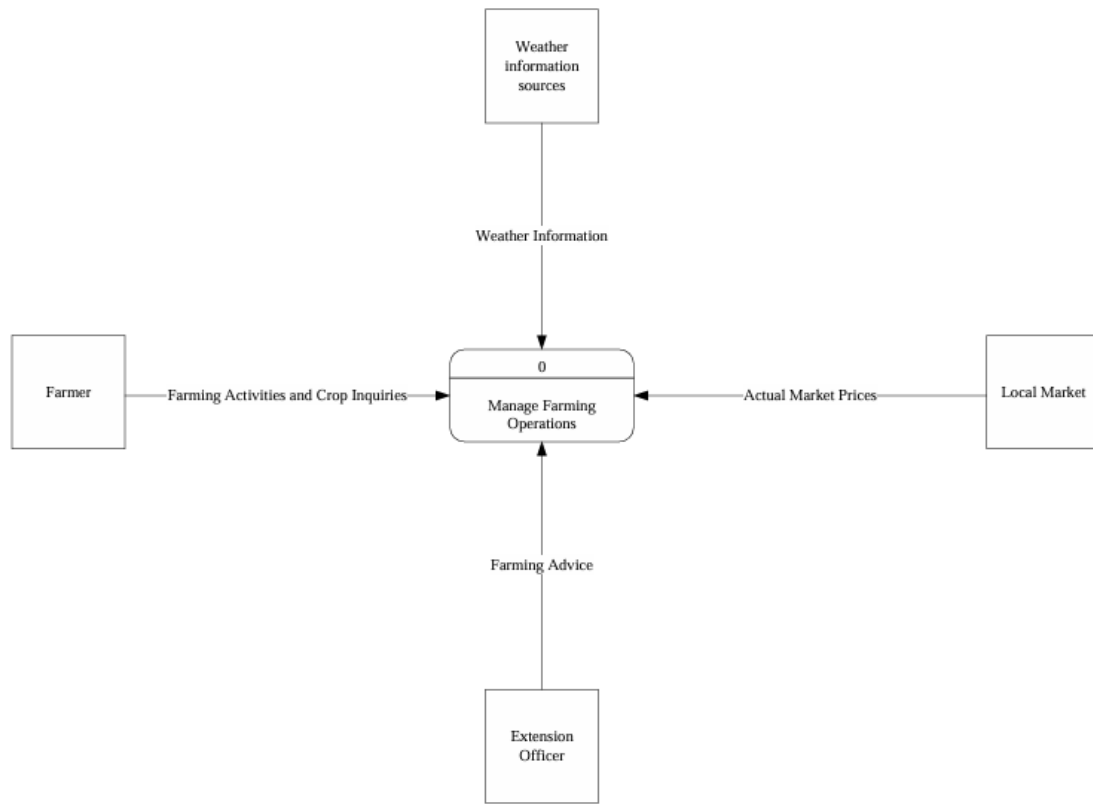
5.0 System Design Overview

The structured design methodology adopted for this project ensures that the system's architecture, components, and interactions are well designed and align with the defined objectives. Structured design is a systematic approach that allows for the decomposition of the system into smaller, manageable modules or components. This method focuses on optimizing the efficiency, maintainability, and scalability of the system, promoting clarity and reusability throughout the design process [3].

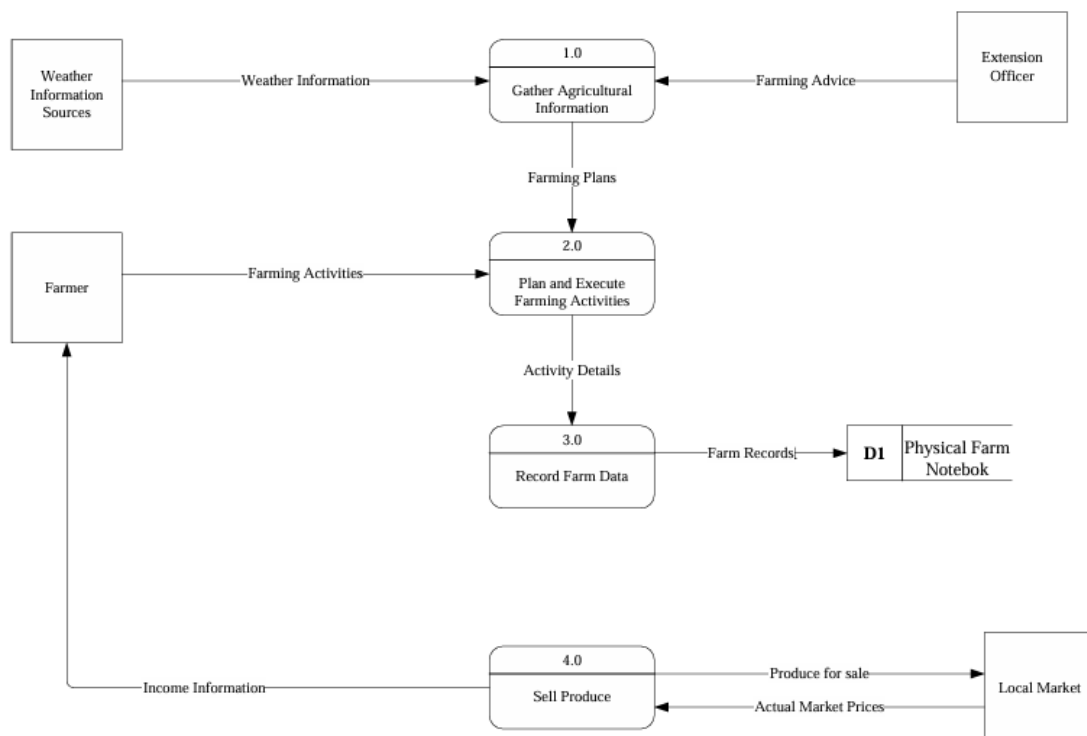
Below, the current system and the proposed system have been outlined in context diagrams and level 0 Data Flow Diagrams.

5.3 DFD diagrams

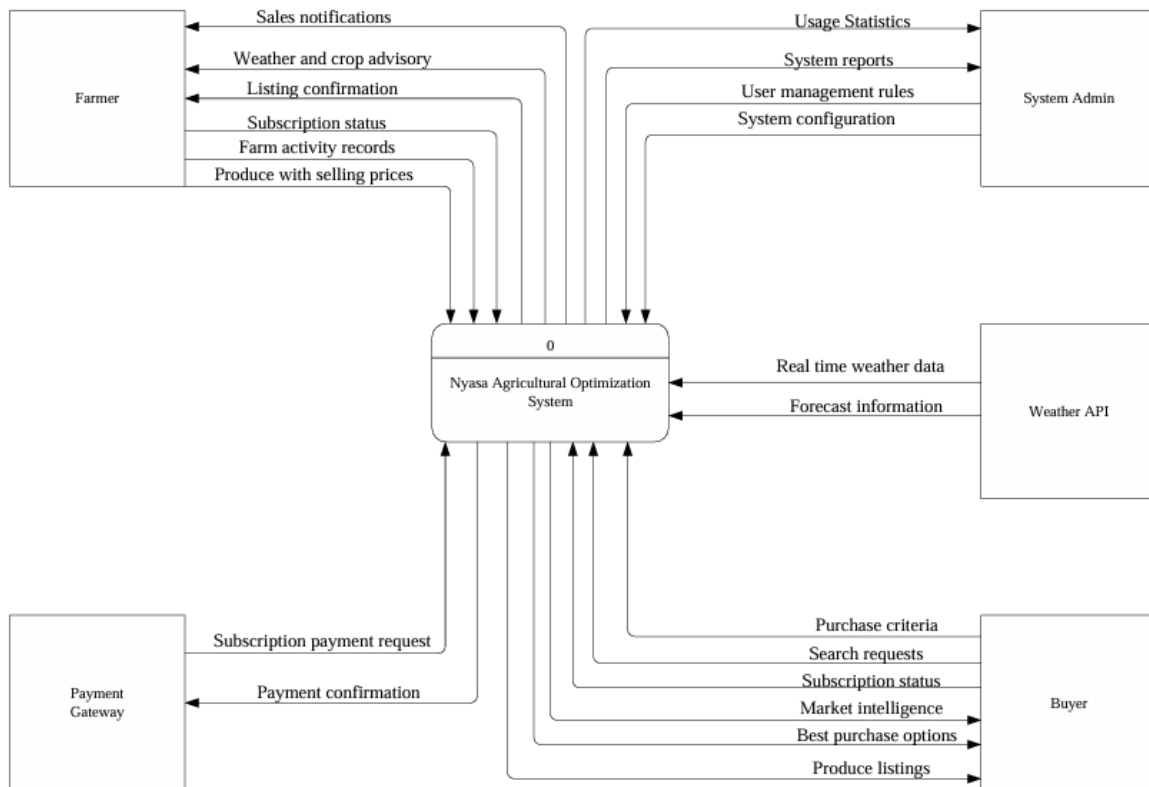
5.3.1 Context Diagram for Current System



5.3.2 Level 0 Diagram for Current System



5.3.3 Context Diagram for Proposed System (NAOS)



5.3.4 Level 0 Diagram for Proposed System (NAOS)

Current System Overview:

Currently, there is no integrated digital system for agricultural optimization in Malawi. Farmers rely on fragmented sources of information and manual processes [2]. The primary actor in all use cases is the Farmer, who manually performs the roles of information gatherer, record-keeper, and decision-maker.

Use Case 1: Obtain Weather Information

- **Actor:** Farmer
- **Description:** The farmer seeks weather forecasts to plan farming activities.
- **Current Flow:** The farmer listens to general regional forecasts on the radio or asks neighbors for recent weather observations [1]. This information is not location-specific and is often unreliable.

Use Case 2: Discover Market Prices

- **Actor:** Farmer
- **Description:** The farmer wants to know current crop prices to decide when and where to sell.
- **Current Flow:** The farmer must physically travel to local markets to inquire with different buyers or rely on word-of-mouth from other farmers, leading to poor bargaining power and exploitative pricing [7].

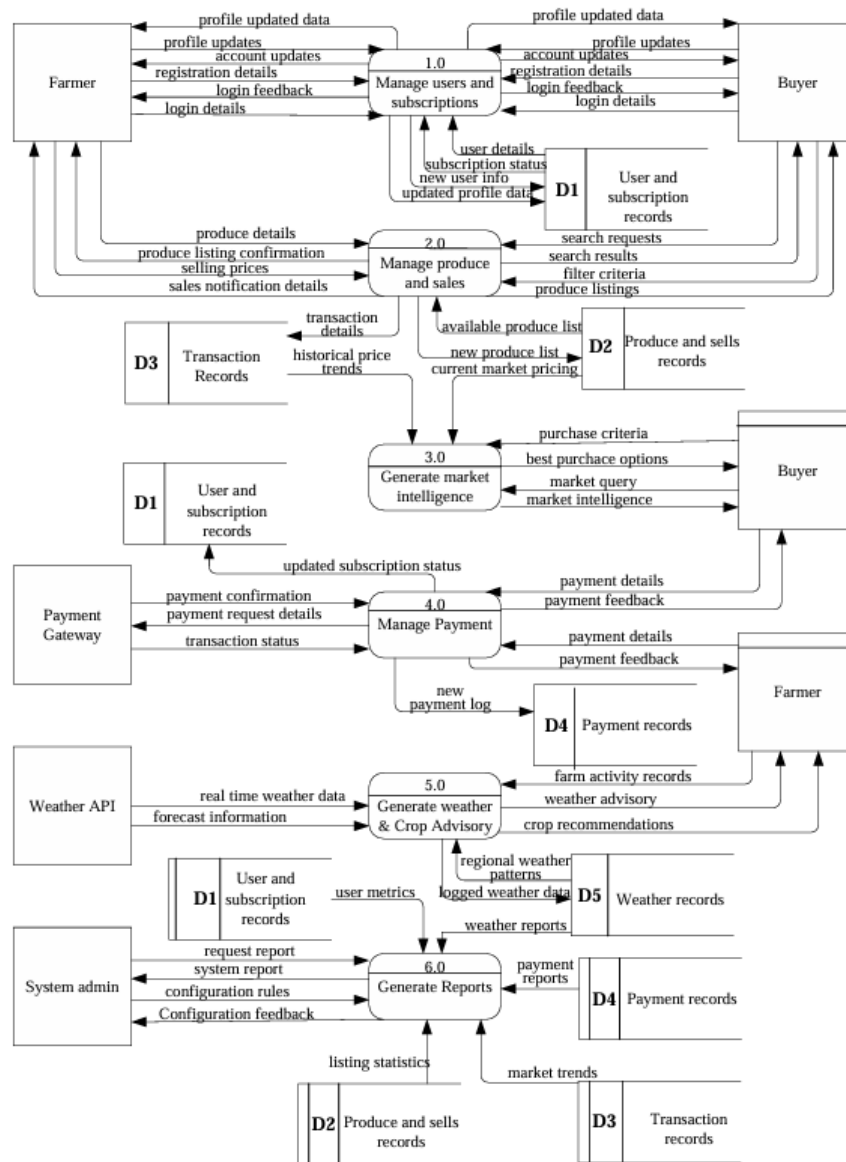
Use Case 3: Record Farm Activities

- **Actor:** Farmer
- **Description:** The farmer needs to keep track of inputs, labor, and harvests.
- **Current Flow:** If done at all, record-keeping is manual in a physical notebook. This makes it difficult to analyze past performance or access historical data for planning [5].

Use Case 4: Make Planting Decisions

- **Actor:** Farmer
- **Description:** The farmer decides what crop to plant and when.
- **Current Flow:** Decisions are based on traditional knowledge, memory of past seasons, and generic advice from extension officers, without data-driven, personalized insights [4].

5.3.2 Context and Level 0 Diagrams for the Proposed System



5.5 Risks

5.5.1 Software Risks

The development and operation of the NAOS software present several risks. A primary risk is dependency on external APIs for weather and market data; if these services become unavailable, change their pricing, or alter their data formats, core system functionalities could fail [9]. Secondly, the accuracy and reliability of the AI recommendation engine is a significant risk. If the algorithm provides incorrect advice due to poor data quality or inadequate model training, it could lead to crop losses and erode user trust [4]. Furthermore, data security and privacy concerns exist, as the system will store sensitive farmer information, making it a target

for breaches [8]. Finally, low digital literacy among the target user base poses an adoption risk, as a complex interface could hinder effective use of the system [2].

5.5.2 Hardware Risks

As a software-based project, NAOS's primary hardware risk concerns the end-user's device ecosystem. Many farmers may have low-end mobile devices with limited processing power, small screens, or outdated operating systems, which could affect the application's performance and usability [10]. Additionally, unreliable internet connectivity in rural Malawi could prevent the system from synchronizing data in real-time or accessing cloud-based services, leading to a poor user experience [6].

6.0 System Requirements

6.1 Functional Requirements

Req. Number	Requirement	Description
1	User Registration	The system shall allow farmers, buyers, and admins to create accounts with role-based access.
2	User Authentication	Users must be able to log in securely using their credentials with authentication protocols.
3	Farm Data Entry	Farmers shall be able to record input usage, yields, and income data in the digital farm diary.
4	Weather Forecast Access	The system shall provide farmers with localized weather forecasts retrieved from the Weather API.
5	Weather Alerts	The system shall notify farmers of extreme weather conditions such as droughts or heavy rainfall.
6	Crop Recommendations	The system shall generate crop recommendations based on weather and historical farm data.
7	Produce Listings	Farmers shall be able to list available produce for sale in the system marketplace.
8	Buyer Inquiries	Buyers shall be able to submit purchase inquiries for listed commodities.

9	Real-Time Price Listings	The system shall provide farmers with real-time commodity price updates from market data feeds.
10	Market Intelligence Reports	The system shall generate market trend reports to guide farmers in decision-making.
11	Farm Record Storage	The system shall store historical farm records for retrieval and analysis by farmers.
12	Usage Reports	The system shall generate usage reports summarizing farmer activities, sales, and system interactions.
13	User Management	System administrators shall be able to add, edit, and deactivate user accounts.
14	System Health Monitoring	The system shall provide administrators with tools to monitor uptime, performance, and errors.
15	Data Export	The system shall allow farmers and admins to export farm and market data into external formats (e.g., PDF, Excel).

6.2 Interface Requirements

Req. Number	Requirement	Description	External System
1	Weather API Integration	The system shall retrieve weather forecasts and alerts automatically.	Weather API
2	Market Data Feed Integration	The system shall update commodity prices and market trends in real time.	Market Data Feed
3	Buyer Portal Access	The system shall allow buyers to interact with farmers via purchase inquiries.	Buyer Portal
4	Produce Listings Visibility	Farmers' produce listings shall be accessible to buyers through the system interface.	Buyer Portal
5	Admin Console Access	The system shall allow administrators to configure system settings and monitor health.	Admin Console

6	Reporting Export	The system shall export usage and market reports to external formats (PDF, Excel).	External Reporting Tools
7	Secure Authentication	The system shall interface with external authentication protocols (e.g., OAuth).	Authentication Service

6.3 Non-Functional Requirements

6.3.1 System-Related Non-Functional Requirements

Req. No.	Non-Functional Requirement	Description	Affected Req No.
1	Performance – Query Speed	The system shall perform simple queries (e.g., produce listings, price checks) in less than 5 seconds.	7, 9
2	Performance – Data Storage	The system shall store up to 10 years of farm records without data loss.	3, 11
3	Reliability	The system shall maintain 99% uptime to ensure farmers and buyers can access services continuously.	1, 2, 7, 9
4	Robustness	The system shall recover gracefully from failures such as API downtime or network interruptions.	4, 5, 9
5	Accuracy of Data	The system shall ensure that market prices and weather forecasts are updated in real time with verified sources.	4, 5, 9
6	Security	The system shall encrypt all user data and transactions using industry-standard protocols (e.g., SSL/TLS).	2, 13
7	Privacy	The system shall restrict access to farm records and personal data to authorized users only.	1, 2, 3, 11
8	Safety	The system shall prevent unauthorized modifications to farm records and market data.	3, 7, 9

9	Portability	The system shall be accessible via web and mobile platforms without loss of functionality.	1, 3, 7, 9
10	Modifiability & Extensibility	The system shall allow future integration with additional APIs (e.g., financial services) without major redesign.	4, 9, 13

6.3.2 User-Related Non-Functional Requirements

Req. No.	Non-Functional Requirement	Description	Affected Req No.
11	Skill Levels	The system shall be usable by farmers with basic digital literacy skills.	1, 3, 7
12	Training	The system shall provide onboarding tutorials and help guides for farmers, buyers, and admins.	1, 2, 13
13	Accessibility	The system shall support multiple languages (including Chichewa and English) to ensure inclusivity.	1, 3, 7
14	Usability	The system shall provide a simple, intuitive interface with minimal navigation steps.	1, 3, 7, 8
15	Support	The system shall provide a helpdesk or support channel for troubleshooting user issues.	1, 2, 13

7.0 Engineering Plan

a. System lifecycle constraints: Development will follow the Agile methodology over an academic-year timeline, constrained by university deadlines for project submission and evaluation [9].

b. System delivery

- I. **Extent of deliverables:** The final deliverables will include the complete source code, database schema, system documentation (SRS, DDD, User Manual), and a deployed, functional prototype.

- II. **Deliverable formats:** Documentation will be in PDF format. Source code will be managed in a Git repository. The application will be delivered as a web application and an Android APK file.

c. Installation

The developer will have initial deployment and configuration access to the production server environment for setup.

d. Scope Management and Reporting: A formal process will be used to manage the project scope. Any proposed change to the requirements in this document must be submitted as a formal Scope Change Request to the Project Owner and Supervisor for approval. Progress against the approved scope will be reported during weekly supervisor meetings and through sprint review demonstrations.

8.0 Glossary

Crop Recommendation: A software component that uses machine learning to analyze data and generate personalized crop and planting advice for farmers [4].

API: Application Programming Interface, a set of rules that allows different software applications to communicate with each other [8].

Digital Farm Diary: A module within NAOS that allows users to digitally record farming activities, inputs, and outputs for better farm management [5].

Market Intelligence: The collection and analysis of information regarding market prices, trends, and buyer information to support informed sales decisions [7].

Smallholder Farmer: A farmer managing a small plot of land, typically relying primarily on family labor and producing for both subsistence and the market [1].

9.0 References

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